

~~[ATTITUDE]~~
[PART II. ATTITUDE]

①

~~< Rigid Body >~~ < Rigid Body >

② Rigid Body $O \subset \mathbb{R}^3$

For $\forall p, q \in O$

$$\|p(t) - q(t)\| = \|p(0) - q(0)\| \quad \text{for } \forall t \geq 0$$

③ Rigid Motion is a continuous motion such that

$$\|p(t) - q(t)\| = \|p(0) - q(0)\| \quad \text{for } \forall t \geq 0.$$

④ A mapping $g: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a Rigid Body Transformation if

(i) Length is preserved

$$\|g(p) - g(q)\| = \|p - q\| \quad \text{for } \forall p, q \in \mathbb{R}^3$$

$$\|p - q\| = \|g(p) - g(q)\|$$

(ii) Cross products are preserved

$$p \times q = g(p) \times g(q) \quad \text{for } \forall p, q \in \mathbb{R}^3$$

(iii) Inner products are preserved

$$p^T q = g^T(p) g(q) \quad \text{for } \forall p, q \in \mathbb{R}^3$$

Ref) Arnold, V.I. : Mathematical Methods of Classical Mechanics

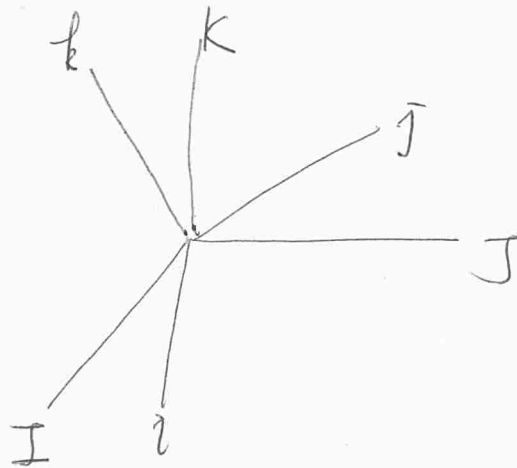
Attitude = Orientation of S/C

SAT 2011

Attitude Representations	No. of Parameters	Characteristics
Direction Cosine Matrix = Rotation Matrix	9	Nonsingular
Euler Principal Angle & Axis	4	Nonsingular
Euler Angles	3	Singular (Gimbal Lock) Physical Interpretation
Quaternion	4	Nonsingular Computational Convenience
Rodrigues Parameters (Gibbs Vector)	3	Singular (180° rotation) Useful for Analytical Study
Modified Rodrigues Parameters	3	Singular (360° rotation) Useful for Analytic Study

 Linked to each other by Mathematical Relations

Rotation Matrix = Direction Cosine Matrix



(Reference Frame IJK (~~is~~) : A
 Body-Fixed Frame ijk (~~is~~) : B
 = rotating

(1) Projection of rotating unit vectors of Frame B onto the fixed unit vectors of Frame A
 = Specification of the coordinates of rotating unit vectors of Frame B in terms of the fixed Frame A

$$\begin{cases} \hat{i} = (i \cdot I) I + (i \cdot J) J + (i \cdot K) K \\ \hat{j} = (j \cdot I) I + (j \cdot J) J + (j \cdot K) K \\ \hat{k} = (k \cdot I) I + (k \cdot J) J + (k \cdot K) K \end{cases}$$

$$\begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} R \triangleq \begin{bmatrix} i \cdot I & i \cdot J & i \cdot K \\ j \cdot I & j \cdot J & j \cdot K \\ k \cdot I & k \cdot J & k \cdot K \end{bmatrix} \Leftarrow \text{Rotation Matrix} \quad \boxed{1}$$

II) Properties of Rotation Matrix

$$① \quad R = [r_1 \ r_2 \ r_3]$$

$$r_i^T r_j = \delta_{ij} \quad \Leftarrow \text{columns \& rows are orthonormal to each other}$$

$$② \quad R R^T = R^T R = I_{3 \times 3}$$

$$③ \quad R^{-1} = R^T$$

④

$$\textcircled{5} \quad \det R = \pm 1 \quad \begin{cases} +1 & \text{for RHS coordinates} \\ -1 & \text{for LHS} \end{cases}$$

④ Set of all rotation matrices has a group structure and is ~~is~~ often denoted by SO(3)
special orthogonality

III) $R \sim$ $\begin{pmatrix} \text{Rotation} \\ \text{Attitude of Rigid Body} \\ \text{Transformation Matrix.} \end{pmatrix}$

IV) Consider a given vector

$$\begin{aligned} \vec{X} &= X\mathbf{i} + Y\mathbf{j} + Z\mathbf{k} \\ &= x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \end{aligned} \quad \Leftarrow \textcircled{\square}$$

$$\Rightarrow \underline{\begin{bmatrix} x \\ y \\ z \end{bmatrix}} = R \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad \Leftrightarrow \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = R^T \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

V) Three Coordinate Frames (A, B, C)

$$\begin{cases} R_{C/B} = \text{rotation matrix of Frame C w.r.t. Frame B} \\ R_{B/A} = \text{" " " " " A} \\ R_{C/A} = \text{" " " " " A} \end{cases}$$

$$R_{C/A} = R_{C/B} R_{B/A}$$

⑥ Euler Principal Angle & Axis

i) Euler's Principal Rotation Theorem

- A rigid body can be brought from an arbitrary initial orientation to an arbitrary orientation by a single rotation through a principal angle (ϕ) and a principal axis ($\vec{a}$)
- $\vec{a}$ is unity by convention.

- Axis / Angle Variables ($\vec{a}, \phi$)
- Exponential Coordinates $\vec{a}\phi$ ($\|\vec{a}\| = 1$)
(= Rotation vector)

ii) Relation to Rotation Matrix

$$\text{Let } S(\vec{a}) \triangleq \frac{1}{\vec{a}} = \vec{a}^\wedge = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

Then, $S(\vec{a}) = -[S(\vec{a})]^T$; Skew-Symmetric

$$\{\vec{a} \times \vec{b}\} = S(\vec{a})\vec{b}$$

~~vector-wise~~

~~only component~~

(6)



$$(\vec{a}, \phi) \Rightarrow R.$$

$$R = e^{S(\vec{a})\phi}$$

$$= I + S(\vec{a})\phi + \frac{1}{2!}[S(\vec{a})\phi]^2 + \dots$$



$$\text{Fact: } S(\vec{a})^2 = \cancel{aa^T} aa^T - \|\vec{a}\|^2 I = aa^T - I$$

$$S(\vec{a})^3 = -\|\vec{a}\|^2 S(\vec{a}) = -S(\vec{a})$$

$$S(\vec{a})^4 = -S(\vec{a})^2$$

$$R = I + S(\vec{a}) \left[\phi - \frac{1}{3!} \phi^3 + \dots \right]$$

$$+ S(\vec{a})^2 \left[\frac{1}{2!} \phi^2 - \frac{1}{4!} \phi^4 + \frac{1}{6!} \phi^6 - \dots \right]$$

$$= I + S(\vec{a}) \sin \phi + S(\vec{a})^2 (1 - \cos \phi)$$

$$\Rightarrow R = e^{S(\vec{a})\phi} = \cos \phi I + aa^T (1 - \cos \phi) + S(\vec{a}) \sin \phi$$

(Rodrigues' Formula)



$$R \Rightarrow (\vec{a}, \phi)$$

$$\phi = \cos^{-1}_{(0, \pi)} \left[\frac{\text{trace}(R) - 1}{2} \right]$$

$$\vec{a} = \frac{1}{2 \sin \phi} \begin{bmatrix} r_{23} - r_{32} \\ r_{31} - r_{13} \\ r_{12} - r_{21} \end{bmatrix} \quad \text{if } \phi \neq 0.$$

$$\vec{a} = \frac{1}{2 \sin \phi} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \quad \text{if } \phi \neq 0$$

T

③ Euler Angles

1) 3 sequential rotations about 3 axes

① 12 combinations

② Gimbal Lock singularity ($\theta = \pm \frac{\pi}{2}$)

✓ ③ Roll rotation

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}$$

✓ ④ Pitch rotation

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$$

✓ ⑤ Yaw rotation

$$R_z(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

ii) Relation to rotation matrix (3-2-1 rotation)

$$① R = R_x(\phi) R_y(\theta) R_z(\psi)$$

$$② R^{-1} = R_z^{-1}(\psi) R_y^{-1}(\theta) R_x^{-1}(\phi)$$

$$= R_z^T(\psi) R_y^T(\theta) R_x^T(\phi)$$

$$= R_z(-\psi) R_y(-\theta) R_x(-\phi)$$

iii) Consider, for example, 3-2-1 rotation, ~~one of the most common rotation sequences~~

① One of the most common rotation sequences.

② Start from ~~an~~ the mertrial (=reference) frame.

rotate sequentially about the intermediate axes

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$$

$$= R(\phi, \theta, \psi) \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= R_z(\phi) R_y(\theta) R_z(\psi) \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

③ Quaternions (= Euler Parameters)

i) 4 scalar quantities (q_1, q_2, q_3, q_4) combined to represent rotations

ii) No singularity & computational convenience

iii) Definition by relation to Euler Principal Angle & Axis
($\vec{a}, \phi$)

$$\textcircled{i} \begin{cases} q_1 = a_1 \sin \frac{\phi}{2} \\ q_2 = a_2 \sin \frac{\phi}{2} \\ q_3 = a_3 \sin \frac{\phi}{2} \\ q_4 = \cos \frac{\phi}{2} \end{cases} \quad \begin{aligned} &\cancel{\sum_{i=1}^4 q_i^2 = 1} \\ &\Rightarrow \sum_{i=1}^4 q_i^2 = 1 \end{aligned}$$

$$\textcircled{ii} \quad \cancel{q = q_4 + q_1 i + q_2 j + q_3 k} \\ \quad \quad \quad = \cancel{q_4 + \vec{q}}$$

$$\textcircled{iii} \quad \cancel{\sum_{i=1}^4 q_i^2 = 1}$$

iv) (q_1, q_2, q_3, q_4) $\rightarrow$ ($\vec{a}, \phi$)

$$\phi = 2 \cos^{-1}(q_4)$$

$$a = \begin{cases} \frac{1}{\sin \frac{\phi}{2}} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}, & \text{if } \phi \neq 0 \\ 0, & \text{if } \phi = 0 \end{cases}$$

V) $(q_1, q_2, q_3, q_4) \Rightarrow R$

$$R = (q_4^2 - \mathbf{q}^T \mathbf{q}) I + 2\mathbf{q}\mathbf{q}^T + 2q_4 S(\mathbf{q})$$

$$\mathbf{q} \triangleq \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}$$

✓ (2) Rodrigues Parameters = Gibbs Vector

~~(1) Useful for~~

i) Definition by relation to $(\vec{a}, \phi)$ and to (q_1, q_2, q_3, q_4)

$$\vec{g} = \vec{a} \tan \frac{\phi}{2} = \frac{\vec{q}}{q_4}$$

ii) Useful for analytical study

iii) Undefined for 180° rotations

Undesirable for global attitude representation

✓ (3) Modified Rodrigues Parameters

i) Definition by relation to $(\vec{a}, \phi)$ and to (q_1, q_2, q_3, q_4)

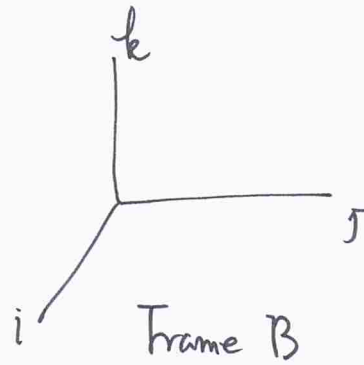
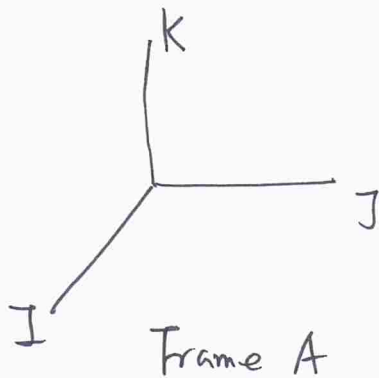
$$\vec{p} = \vec{a} \tan \frac{\phi}{4} = \frac{\vec{q}}{1 + q_4}$$

ii) Useful for analytical study

Undefined for 360° rotations

ATTITUDE KINEMATICS

① Kinematics of Rotating Frame (itself)



$$\vec{\omega} = \vec{\omega}_{B/A} = \omega_x i + \omega_y j + \omega_z k$$

= angular velocity of Frame B w.r.t. Frame A

• Time derivatives of the unit vectors (i, j, k) w.r.t. Frame A

$$\begin{cases} \frac{di}{dt} = \vec{\omega} \times i \Leftrightarrow S(\vec{\omega}) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{di}{dt} \\ 0 \\ 0 \end{bmatrix} \\ \frac{dj}{dt} = \vec{\omega} \times j \Leftrightarrow S(\vec{\omega}) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{dj}{dt} \\ 0 \end{bmatrix} \\ \frac{dk}{dt} = \vec{\omega} \times k \Leftrightarrow S(\vec{\omega}) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{dk}{dt} \end{bmatrix} \end{cases}$$

Perpendicular to the plane generated by $\vec{\omega} = \vec{\omega}_{B/A}$ and the unit vectors (i, j, k) respectively

$$\text{cf. } \begin{bmatrix} \frac{di}{dt} \\ \frac{dj}{dt} \\ \frac{dk}{dt} \end{bmatrix} \neq S(\vec{\omega}) \begin{bmatrix} i \\ j \\ k \end{bmatrix}, \quad \begin{bmatrix} \frac{di}{dt} \\ \frac{dj}{dt} \\ \frac{dk}{dt} \end{bmatrix} =$$

$$\begin{aligned} \hookrightarrow &= \vec{\omega} \times \begin{bmatrix} i \\ j \\ k \end{bmatrix} \Leftrightarrow S(\vec{\omega}) \begin{bmatrix} \{i\} \\ \{j\} \\ \{k\} \end{bmatrix} = S(\vec{\omega}) \begin{bmatrix} [1 \ 0 \ 0] \\ [0 \ 1 \ 0] \\ [0 \ 0 \ 1] \end{bmatrix} \end{aligned}$$

① Kinematics in the Rotating Frame

$$\vec{r} = r_1 \vec{i} + r_2 \vec{j} + r_3 \vec{k}$$

$$\left(\frac{d\vec{r}}{dt}\right)_A = (\dot{r}_1 \vec{i} + \dot{r}_2 \vec{j} + \dot{r}_3 \vec{k}) + (r_1 \dot{\vec{i}} + r_2 \dot{\vec{j}} + r_3 \dot{\vec{k}})$$

$$= \left(\frac{d\vec{r}}{dt}\right)_B + [r_1(\vec{\omega} \times \vec{i}) + r_2(\vec{\omega} \times \vec{j}) + r_3(\vec{\omega} \times \vec{k})]$$

$$= \left(\frac{d\vec{r}}{dt}\right)_B + [(\vec{\omega} \times r_1 \vec{i}) + (\vec{\omega} \times r_2 \vec{j}) + (\vec{\omega} \times r_3 \vec{k})]$$

$$= \left(\frac{d\vec{r}}{dt}\right)_B + \vec{\omega} \times (r_1 \vec{i} + r_2 \vec{j} + r_3 \vec{k})$$

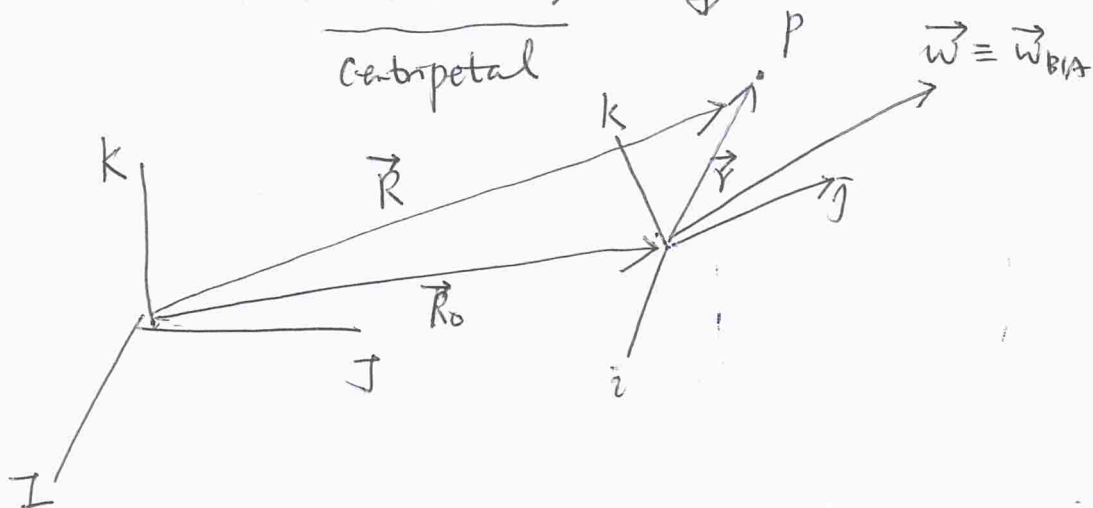
$$\Rightarrow \boxed{\left(\frac{d\vec{r}}{dt}\right)_A = \left(\frac{d\vec{r}}{dt}\right)_B + \vec{\omega} \times \vec{r}} \quad (2)$$

$$\vec{R} = \vec{R}_0 + \vec{r}$$

$$\left(\frac{d\vec{R}}{dt}\right)_A = \left(\frac{d\vec{R}_0}{dt}\right)_A + \left(\frac{d\vec{r}}{dt}\right)_A = \left(\frac{d\vec{R}_0}{dt}\right)_A + \cancel{\frac{d\vec{r}}{dt}} \left(\frac{d\vec{r}}{dt}\right)_B + \vec{\omega} \times \vec{r} \quad (3)$$

$$\left(\frac{d\vec{R}}{dt}\right)_A = \left(\frac{d^2\vec{R}_0}{dt^2}\right)_A + \left(\frac{d^2\vec{r}}{dt^2}\right)_B + \underbrace{\vec{\omega} \times \vec{r}}_{\text{tangential}} + \underbrace{2\vec{\omega} \times \left(\frac{d\vec{r}}{dt}\right)_B}_{\text{Coriolis}}$$

$$+ \underbrace{\vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{Centripetal}} \quad (4)$$



① Attitude Kinematics 1 : Rotation (= Direction Cosine) Matrix

$$\circ \vec{\omega} = \vec{\omega}_{B/A}^B = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \Rightarrow S(\vec{\omega}) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

$$\circ \begin{bmatrix} i \\ j \\ k \end{bmatrix} = R \begin{bmatrix} I \\ J \\ K \end{bmatrix}$$

$$\Rightarrow \frac{d}{dt} \begin{bmatrix} i \\ j \\ k \end{bmatrix} = \dot{R} \begin{bmatrix} I \\ J \\ K \end{bmatrix} \quad (4) \quad (\text{direct differentiation})$$

$$\frac{d}{dt} \begin{bmatrix} i \\ j \\ k \end{bmatrix} = \vec{\omega} \times \begin{bmatrix} i \\ j \\ k \end{bmatrix} = \vec{\omega} \times R \begin{bmatrix} I \\ J \\ K \end{bmatrix} = S(\vec{\omega}) R \begin{bmatrix} I \\ J \\ K \end{bmatrix} \quad (5) \quad (\text{attitude kinematics})$$

$$\Rightarrow (4) = (5)$$

$$\dot{R} = S(\vec{\omega}) R \quad (6)$$

◦ No singularity

② Attitude Kinematics 2 : Euler Principal Angle & Axis ($\vec{a}, \phi$)

$$\dot{\phi} = \vec{a}^T \vec{\omega}$$

$$\dot{\vec{a}} = \frac{1}{2} S(\vec{\omega}) \vec{a} + \frac{1}{2} \cot\left(\frac{\phi}{2}\right) [\vec{\omega} - (\vec{\omega}^T \vec{a}) \vec{a}]$$

◦ Singularity at $\phi = 0, \pm\pi$.

For an arbitrary fixed vector X in the inertial frame

$$\{x(t)\} = R(t) \{X\}$$

$$\Rightarrow \{\dot{x}(t)\} = \dot{R}(t) \{X\} \quad (4)$$

$$\{\dot{x}(t)\} = S(\vec{\omega}) \{x(t)\} = S(\vec{\omega}) R(t) \{X\} \quad (5)$$

$$\Rightarrow (4) = (5), \quad \dot{R} = S(\vec{\omega}) R$$

Attitude Kinematics 3: Euler Angles

1) Consider 3-2-1 rotation

$$\begin{cases} R_z(\psi) : A \rightarrow A' \\ R_y(\theta) : A' \rightarrow A'' \\ R_x(\phi) : A'' \rightarrow B \end{cases} \Rightarrow \begin{cases} \vec{\omega}_{A/A} = \dot{\psi} \mathbf{k} = \dot{\psi} \mathbf{k}' \\ \vec{\omega}_{A''/A'} = \dot{\theta} \mathbf{j}' = \dot{\theta} \mathbf{j}'' \\ \vec{\omega}_{B/A''} = \dot{\phi} \mathbf{i}'' = \dot{\phi} \mathbf{i} \end{cases}$$

$$\vec{\omega}_{B/A} = \vec{\omega}_{B/A''} + \vec{\omega}_{A''/A'} + \vec{\omega}_{A'/A}$$

$$= \dot{\phi} \mathbf{i} + \dot{\theta} \mathbf{j}'' + \dot{\psi} \mathbf{k}'$$

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} + R_x(\phi) \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + R_x(\phi) R_y(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \sin\phi \cos\theta \\ 0 & -\sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \frac{1}{\cos\theta} \begin{bmatrix} \cos\theta & \sin\phi \sin\theta & \cos\phi \sin\theta \\ 0 & \cos\phi \cos\theta & -\sin\phi \cos\theta \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

ii) Singularity at $\theta = \pm \frac{\pi}{2}$



Attitude Kinematics 4: Quaternion

i) Start from the kinematics of Rotation Matrix

$$\dot{R} = S(\vec{\omega})R$$

ii) Introduce Rotation Matrix $\Leftrightarrow$ Quaternion relation

$$\begin{cases} R = (q_4^2 - \mathbf{q}^T \mathbf{q}) \mathbf{I} + 2\mathbf{q}\mathbf{q}^T + 2q_4 S(\mathbf{q}) \\ \mathbf{q} \triangleq [q_1 \ q_2 \ q_3]^T \end{cases}$$

results in:

$$\omega_x = 2(\dot{q}_1 q_4 + \dot{q}_2 q_3 - \dot{q}_3 q_2 - \dot{q}_4 q_1) \quad (1)$$

$$\omega_y = 2(\dot{q}_2 q_4 + \dot{q}_3 q_1 - \dot{q}_1 q_3 - \dot{q}_4 q_2) \quad (2)$$

$$\omega_z = 2(\dot{q}_3 q_4 + \dot{q}_1 q_2 - \dot{q}_2 q_1 - \dot{q}_4 q_3) \quad (3)$$

iii) Differentiate the identity

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1$$

$$\Rightarrow 2(\dot{q}_1 q_1 + \dot{q}_2 q_2 + \dot{q}_3 q_3 + \dot{q}_4 q_4) = 0 \quad (4)$$

iv) Group 4 scalar ~~equations~~ diff. equations (1) ~ (4)

$$\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \\ 0 \end{bmatrix} = 2 \begin{bmatrix} q_4 & q_3 & -q_2 & -q_1 \\ -q_3 & q_4 & q_1 & -q_2 \\ q_2 & -q_1 & q_4 & -q_3 \\ q_1 & q_2 & q_3 & q_4 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & -q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{bmatrix} w_x \\ w_y \\ w_z \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & w_z & -w_y & w_x \\ -w_z & 0 & w_x & w_y \\ w_y & -w_x & 0 & w_z \\ -w_x & -w_y & -w_z & 0 \end{bmatrix}$$

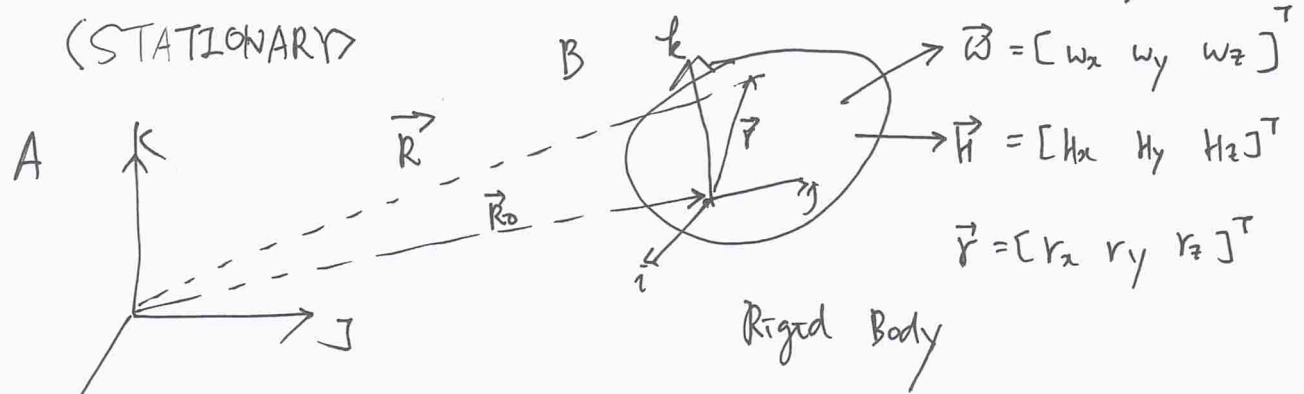
$$\Rightarrow \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & w_z & -w_y & w_x \\ -w_z & 0 & w_x & w_y \\ w_y & -w_x & 0 & w_z \\ -w_z & -w_y & -w_z & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} S(\vec{w}) \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \frac{1}{2} q_4 \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$$

$$\dot{q}_4 = -\frac{1}{2} \vec{w}^T \vec{q}$$

< ATTITUDE DYNAMICS & STABILITY ANALYSIS >

(STATIONARY)



② Angular Momentum

$$i) \vec{H} \triangleq \int \vec{r} \times (\vec{\omega} \times \vec{r}) dm$$

$$= \int_M [\omega_x (r_y^2 + r_z^2) - \omega_y r_x r_y - \omega_z r_x r_z] dm \vec{i} \quad (H_x)$$

$$+ \int_M [\omega_y (r_z^2 + r_x^2) - \omega_z r_y r_z - \omega_x r_x r_y] dm \vec{j} \quad (H_y)$$

$$+ \int_M [\omega_z (r_x^2 + r_y^2) - \omega_x r_x r_z - \omega_y r_y r_z] dm \vec{k} \quad (H_z)$$

ii) Moment of Inertia

$$I_x = \int_M (r_y^2 + r_z^2) dm$$

$$I_y = \int_M (r_z^2 + r_x^2) dm$$

$$I_z = \int_M (r_x^2 + r_y^2) dm$$

$$I_{xy} = - \int_M r_x r_y dm$$

$$I_{yz} = - \int_M r_y r_z dm$$

$$I_{zx} = - \int_M r_z r_x dm$$

$$I_x + I_y > I_z$$

$$I_y + I_z > I_x$$

$$I_z + I_x > I_y$$

ii) (i) & (ii)

$$\vec{H} = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ I_{yx} & I_y & I_{yz} \\ I_{zx} & I_{zy} & I_z \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \mathbf{I} \vec{\omega}$$

③ Newton's Second Law of Motion

i) (Time-rate of change of Angular Momentum)

= (Sum of all external moments acted on rigid body)

$$ii) \left(\frac{d\vec{H}}{dt} \right)_A = \left(\frac{d\vec{H}}{dt} \right)_B + \vec{\omega} \times \vec{H} = \vec{M}$$

$$\Rightarrow \Rightarrow \boxed{\mathbf{I} \dot{\vec{\omega}} + \vec{\omega} \times (\mathbf{I} \vec{\omega}) = \vec{M}}$$

(Euler's Rotational Eqs of Motion)

$$iii) \frac{d}{dt} \left(\frac{I_A \omega_{B/A}^A}{I_A \omega_{B/A}^A} \right) = \vec{M}_A \quad A: \text{Inertial Frame}$$

$$\begin{cases} \vec{\omega}_A^A = R^T \vec{\omega}_{B/A}^B, & R = R_{B/A} = R_{AB}^T \\ I_A = R^T I_B R \end{cases}$$

I_A : variable. I_B : constant (in general)

$$\Rightarrow \frac{d}{dt} \left(\underbrace{R^T I_B R}_{\mathbf{I}_{3 \times 3}} \vec{\omega}_{B/A}^B \right) = \vec{M}_A$$

$$\Rightarrow \dot{R}^T I_B \vec{\omega}_{B/A}^B + R^T I_B \dot{\vec{\omega}}_{B/A}^B = \vec{M}_A$$

$$\Rightarrow R^T S(\vec{\omega}_{B/A}^B) I_B \vec{\omega}_{B/A}^B + R^T I_B \dot{\vec{\omega}}_{B/A}^B = \vec{M}_A$$

$$\Rightarrow S(\vec{\omega}_B) I_B \vec{\omega}_B + I_B \dot{\vec{\omega}}_B = R \vec{M}_A$$

$$\Rightarrow \boxed{I_B \dot{\vec{\omega}}_B + \vec{\omega}_B \times (I_B \vec{\omega}_B) = \vec{M}_B}$$

$$IV) \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ & I_y & I_{yz} \\ & & I_z \end{bmatrix} \begin{bmatrix} \dot{w}_x \\ \dot{w}_y \\ \dot{w}_z \end{bmatrix} + \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix} \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ & I_y & I_{yz} \\ & & I_z \end{bmatrix} \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix}$$

V) If Body-fixed frame has the center of mass at origin and aligned with principal axes, then

$$I_B = \text{diag}(I_x, I_y, I_z)$$

Euler's rotational eqns of motion is reduced to

$$\begin{cases} I_x \dot{w}_x = (I_y - I_z) w_y w_z + M_x \\ I_y \dot{w}_y = (I_z - I_x) w_z w_x + M_y \\ I_z \dot{w}_z = (I_x - I_y) w_x w_y + M_z \end{cases}$$

© Rotational Kinematics & Dynamics Combined.

$$I_B \dot{w}_B + w_B \times (I_B w_B) = M_B$$

$$\dot{R} = S(w_B) R \quad (\text{for Euler Angle, Quaternion, (2, \phi) kinematics})$$

(or Euler Angle, Quaternion, Euler Principal Angle & Axis Kinematics)

② Torque-free Motion ($M_x = M_y = M_z = 0$)

$$\cancel{I_x \dot{w}_x} \quad I_B \dot{w}_B + w_B \times (I_B w_B) = 0$$

$$\Rightarrow \begin{cases} I_x \dot{w}_x - (I_y - I_z) w_y w_z = 0 \\ I_y \dot{w}_y - (I_z - I_x) w_x w_z = 0 \\ I_z \dot{w}_z - (I_x - I_y) w_x w_y = 0 \end{cases}$$

~~I) An Equilibrium~~

I) Equilibrium

$$w_B \times I_B w_B = 0 \quad \Leftrightarrow \quad w_B \parallel I_B w_B$$

II) Conservation of (Kinetic) Energy

$$T = \frac{1}{2} w_B^T I_B w_B$$

$$= \frac{1}{2} (I_x w_x^2 + I_y w_y^2 + I_z w_z^2)$$

$$\dot{T} = I_x w_x \dot{w}_x + I_y w_y \dot{w}_y + I_z w_z \dot{w}_z$$

$$= w_x [(I_y - I_z) w_y w_z] + w_y [(I_z - I_x) w_x w_z] + \dots$$

$$= 0$$

$$\Rightarrow T = \text{constant}$$

III) Conservation of Magnitude of Angular Momentum

$$H^2 = |I_B w_B|^2 = (I_B w_B)^T (I_B w_B)$$

$$= I_x^2 w_x^2 + I_y^2 w_y^2 + I_z^2 w_z^2$$

$$\Rightarrow 2H\dot{H} = 2I_x^2 w_x \dot{w}_x + 2I_y^2 w_y \dot{w}_y + 2I_z^2 w_z \dot{w}_z$$

$$\Rightarrow H\dot{H} = I_x w_x [(I_y - I_z) w_y w_z] + I_y w_y [(I_z - I_x) w_x w_z] + \dots = 0$$

$$\Rightarrow \dot{H} = \text{constant}$$

In general, angular momentum with respect to the body-fixed frame is not conserved even if $\vec{M}_B = 0$.

© Equilibrium & Linearization for Special Case. (Torque-free Motion)

i) Axisymmetric Spacecraft. ($\Leftrightarrow I_x = I_y \neq I_z$.)

$$\begin{cases} I \dot{\omega}_x = (I - I_z) \omega_y \omega_z \\ I \dot{\omega}_y = (I_z - I) \omega_x \omega_z \\ I_z \dot{\omega}_z = 0 \end{cases}$$

Equilibrium $\omega_x = \omega_y = 0, \quad \omega_z = n$.

Linearization w.r.t. the equilibrium

$$\begin{cases} I \Delta \dot{\omega}_x = (I - I_z) n \Delta \omega_y \\ I \Delta \dot{\omega}_y = -(I - I_z) n \Delta \omega_x \\ I_z \Delta \dot{\omega}_z = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \Delta \dot{\omega}_x = \lambda \Delta \omega_y \\ \Delta \dot{\omega}_y = -\lambda \Delta \omega_x \\ \Delta \dot{\omega}_z = 0 \end{cases} \quad \lambda = \frac{I - I_z}{I}$$

Eigenvalue

$$\begin{pmatrix} s^2 + \lambda^2 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} s = \pm j|\lambda| \\ 0 \end{pmatrix}$$

ii) Fully Symmetric Spacecraft ($I = I_x = I_y = I_z$)

Eqm

$$\begin{pmatrix} I \dot{w}_x = 0 \\ I \dot{w}_y = 0 \\ I \dot{w}_z = 0 \end{pmatrix} \Rightarrow \begin{pmatrix} w_x \\ w_y \\ w_z \end{pmatrix} = \text{constat}$$

Equilibrium

$$\begin{pmatrix} w_x = \bar{w}_x \\ w_y = \bar{w}_y \\ w_z = \bar{w}_z \end{pmatrix} = \text{constat}$$

Angular Momentum is conserved for this special case.

Linearization w.r.t. the equilibrium

$$\begin{pmatrix} I (\dot{\bar{w}}_x + \Delta \dot{w}_x) = 0 \\ I (\dot{\bar{w}}_y + \Delta \dot{w}_y) = 0 \\ I (\dot{\bar{w}}_z + \Delta \dot{w}_z) = 0 \end{pmatrix} \Rightarrow \begin{pmatrix} I \Delta \dot{w}_x = 0 \\ I \Delta \dot{w}_y = 0 \\ I \Delta \dot{w}_z = 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} I \Delta \dot{w}_x = 0 \\ I \Delta \dot{w}_y = 0 \\ I \Delta \dot{w}_z = 0 \end{pmatrix}$$

Eigenvalue

$$s = 0, 0, 0.$$

iii) Asymmetric Spacecraft ($I_x \neq I_y \neq I_z$)

① Equilibrium

$$w_x = w_y = w_z = 0 \quad (\text{rest})$$

Linearization

$$\begin{pmatrix} I_x \dot{w}_x = 0 \\ I_y \dot{w}_y = 0 \\ I_z \dot{w}_z = 0 \end{pmatrix} \Rightarrow \begin{pmatrix} I_x \Delta \dot{w}_x = 0 \\ I_y \Delta \dot{w}_y = 0 \\ I_z \Delta \dot{w}_z = 0 \end{pmatrix}$$

Eigenvalue

$$s = 0, 0, 0$$

② Equilibrium

$$\begin{cases} \omega_x = \bar{\omega}_x, & \omega_y = \omega_z = 0 \\ \omega_y = \bar{\omega}_y, & \omega_x = \omega_z = 0 \\ \omega_z = \bar{\omega}_z, & \omega_x = \omega_y = 0 \end{cases}$$

Consider the case of $\omega_z = \bar{\omega}_z$, $\omega_x = \omega_y = 0$

Linearization w.r.t. the equilibrium

$$\begin{cases} \omega_x = 0 + \Delta\omega_x \\ \omega_y = 0 + \Delta\omega_y \\ \omega_z = \bar{\omega}_z + \Delta\omega_z \end{cases}$$

$$\Rightarrow \begin{cases} \Delta\dot{\omega}_x = \left[\frac{I_y - I_z}{I_x} \bar{\omega}_z \right] \Delta\omega_y \\ \Delta\dot{\omega}_y = \left[\frac{I_z - I_x}{I_y} \bar{\omega}_z \right] \Delta\omega_x \\ \Delta\dot{\omega}_z = 0 \end{cases}$$

Char. Eqn for the first two eqns

$$s^2 + \bar{\omega}_z^2 \frac{(I_y - I_z)(I_x - I_z)}{I_x I_y} = 0$$

Eigenvalues are:

If $(I_x < I_z, I_y < I_z)$ or $(I_z < I_x, I_z < I_y)$,
 I_z is major I_z is minor

then $s = 0, \pm j \bar{\omega}_z \sqrt{\frac{(I_y - I_z)(I_x - I_z)}{I_x I_y}}$

If $(I_x < I_z, I_y > I_z)$ or $(I_x > I_z, I_y < I_z)$
 I_z is intermediate.

then $s = 0, \pm \bar{\omega}_z \sqrt{\frac{(I_y - I_z)(I_x - I_z)}{I_x I_y}}$

② Kinematics & Dynamics combined

$$\begin{cases}
 I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x \\
 I_y \dot{\omega}_y = (I_z - I_x) \omega_z \omega_x + M_y \\
 I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z \\
 \dot{\psi} = (\omega_y \sin \phi + \omega_z \cos \phi) \sec \theta \\
 \dot{\theta} = \omega_y \cos \phi - \omega_z \sin \phi \\
 \dot{\phi} = \omega_x + \omega_y \sin \phi \tan \theta + \omega_z \cos \phi \tan \theta
 \end{cases}$$

T) Perturbations near 'rest' solution

Let $M_x = M_y = M_z \equiv 0$ (torque free)

Then, $(\omega_x = \omega_y = \omega_z = 0$ is a set of equilibrium
 ψ, θ, ϕ are arbitrary constants

Let $\psi = \theta = \phi = 0$ w.l.o.g.

Define

$$\begin{cases}
 \omega_x = 0 + \Delta \omega_x \\
 \omega_y = 0 + \Delta \omega_y \\
 \omega_z = 0 + \Delta \omega_z \\
 \psi = 0 + \Delta \psi \\
 \theta = 0 + \Delta \theta \\
 \phi = 0 + \Delta \phi
 \end{cases}$$

Linearization

$$\begin{cases}
 I_x \Delta \dot{\omega}_x = M_x \\
 I_y \Delta \dot{\omega}_y = M_y \\
 I_z \Delta \dot{\omega}_z = M_z \\
 \Delta \dot{\psi} = \Delta \omega_z \\
 \Delta \dot{\theta} = \Delta \omega_y \\
 \Delta \dot{\phi} = \Delta \omega_x
 \end{cases}
 \Rightarrow
 \begin{cases}
 I_x \Delta \ddot{\phi} = M_x \\
 I_y \Delta \ddot{\theta} = M_y \\
 I_z \Delta \ddot{\psi} = M_z
 \end{cases}$$

ii) Perturbations near 'pure spin' solution

Let $M_x = M_y = M_z = 0$ (torque free)

Then $\omega_x = \omega_y = 0, \quad \omega_z = \bar{\omega}_z = \text{const}$
 $\phi = \theta = 0, \quad \psi = \bar{\omega}_z t$ is a ~~set~~ solution

Define

$$\begin{cases} \omega_x = 0 + \Delta\omega_x \\ \omega_y = 0 + \Delta\omega_y \\ \omega_z = \bar{\omega}_z + \Delta\omega_z \\ \phi = 0 + \Delta\phi \\ \theta = 0 + \Delta\theta \\ \psi = \bar{\omega}_z t + \Delta\psi \end{cases}$$

Linearization

$$\begin{cases} I_x \Delta\dot{\omega}_x = [(I_y - I_z) \bar{\omega}_z] \Delta\omega_y + M_x \\ I_y \Delta\dot{\omega}_y = [(I_z - I_x) \bar{\omega}_z] \Delta\omega_x + M_y \\ I_z \Delta\dot{\omega}_z = M_z \\ \cancel{\bar{\omega}_z} + \Delta\dot{\psi} = \cancel{\bar{\omega}_z} + \Delta\omega_z \end{cases} \text{Decoupled Yaw dynamics}$$

$$\begin{aligned} \Delta\ddot{\theta} &= \Delta\omega_y - \bar{\omega}_z \Delta\phi \\ \Delta\dot{\phi} &= \Delta\omega_x + \bar{\omega}_z \Delta\theta \end{aligned}$$

$$\begin{cases} I_x [\Delta\ddot{\phi} - \bar{\omega}_z \Delta\dot{\theta}] = [(I_y - I_z) \bar{\omega}_z] (\Delta\dot{\theta} + \bar{\omega}_z \Delta\phi) + M_x \\ I_y [\Delta\ddot{\theta} + \bar{\omega}_z \Delta\dot{\phi}] = [(I_z - I_x) \bar{\omega}_z] (\Delta\dot{\phi} - \bar{\omega}_z \Delta\theta) + M_y \\ I_z \Delta\ddot{\psi} = M_z \end{cases}$$

Coupling between roll & pitch dynamics

$$\begin{bmatrix} I_x & \\ & I_y \end{bmatrix} \begin{bmatrix} \Delta \ddot{\phi} \\ \Delta \ddot{\theta} \end{bmatrix} + \begin{bmatrix} 0 & \bar{\omega}_z (-I_x - I_y + I_z) \\ \bar{\omega}_z (I_y - I_z + I_x) & 0 \end{bmatrix} \begin{bmatrix} \Delta \dot{\phi} \\ \Delta \dot{\theta} \end{bmatrix} + \begin{bmatrix} \bar{\omega}_z^2 (I_z - I_y) & 0 \\ 0 & \bar{\omega}_z^2 (I_z - I_x) \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta \theta \end{bmatrix} = \begin{bmatrix} M_x \\ M_y \end{bmatrix}$$

E-values of yaw - eqns, $s=0, 0$. unstable

" roll-pitch eqns $\Rightarrow$ 4 eigenvalues.

< ATTITUDE DYNAMICS & STABILITY ANALYSIS >

(On - Orbit)

⊙ Assumptions

- i) Inertial (= Reference, A), LVLH (L), and Body-fixed frames ($\{b\}$) are constructed
- ii) S/C center of mass is in a circular orbit of radius R and its associated (constant) angular velocity of $\omega_0 = \sqrt{\frac{\mu}{R^3}}$
- iii) S/C is rigid
- iv) S/C Body-fixed axes are principal
- v) Gravity Gradient Moments on the S/C are included.

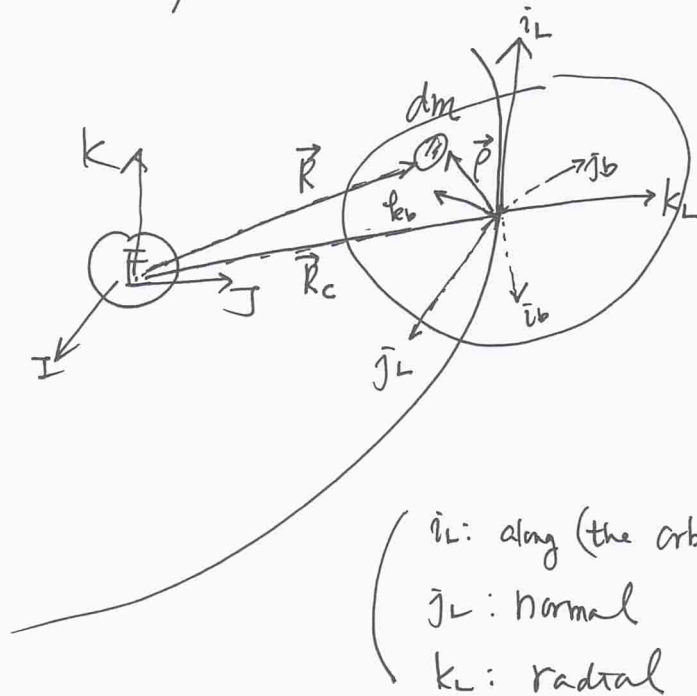
⊙ Variables Definition

- i) $(\omega_x, \omega_y, \omega_z)$ are the components of the angular velocity vector of the S/C w.r.t. the inertial frame, expressed in the body-fixed frame
- ii) $(\dot{\omega}_x^r, \dot{\omega}_y^r, \dot{\omega}_z^r)$ are the components of the 'relative' angular velocity vector of the S/C w.r.t. the LVLH frame, expressed in the body-fixed frame
- iii) (ψ, θ, ϕ) are 3-2-1 Euler Angles that define the S/C attitude w.r.t. the LVLH frame
- iv) (M_x^g, M_y^g, M_z^g) are the components of gravity gradient moments on the S/C in the body-fixed frame
- v) (M_x, M_y, M_z) are the components of the S/C moment vector ...

... in the body-fixed frame due to propulsive/non-gravitational forces

① Effect of Gravitational Forces/Moments

First considered by D'Alembert & Euler



(i_L: along (the orbital velocity vector)
j_L: normal
k_L: radial

$$d\vec{f} = - \frac{\mu \vec{r} dm}{|\vec{r}|^3} = - \frac{\mu (\vec{r}_c + \vec{r}) dm}{|\vec{r}_c + \vec{r}|^3}$$

$$\begin{aligned} \vec{M} &= \int \vec{r} \times d\vec{f} = -\mu \int \frac{\vec{r} \times (\vec{r}_c + \vec{r})}{|\vec{r}_c + \vec{r}|^3} dm \\ &= 3 \omega^2 \vec{r}_c \times (I \vec{r}_c) \end{aligned}$$

$$\begin{bmatrix} M_x^G \\ M_y^G \\ M_z^G \end{bmatrix} = \begin{bmatrix} -3 \omega^2 (I_y - I_z) \sin \phi \cos \phi \cos^2 \theta \\ 3 \omega^2 (I_z - I_x) \cos \phi \sin \theta \cos \theta \\ 3 \omega^2 (I_x - I_y) \sin \phi \sin \theta \cos \theta \end{bmatrix}$$

© On-Orbit Rotational EOM

$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x^G + M_x \\ I_y \dot{\omega}_y = (I_z - I_x) \omega_z \omega_x + M_y^G + M_y \\ I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z^G + M_z \end{cases}$$

$$\begin{cases} \dot{\psi} = (\omega_y^r \sin \phi + \omega_z^r \cos \phi) \sec \theta \\ \dot{\theta} = \omega_y^r \cos \phi - \omega_z^r \sin \phi \\ \dot{\phi} = \omega_x^r + \omega_y^r \sin \phi \tan \theta + \omega_z^r \cos \phi \tan \theta \end{cases}$$

$$\begin{cases} \omega_x^r = \omega_x - \omega_0 \cos \theta \sin \psi \\ \omega_y^r = \omega_y - \omega_0 (\cos \phi \cos \psi + \sin \phi \sin \theta \sin \psi) \\ \omega_z^r = \omega_z - \omega_0 (-\sin \phi \cos \psi + \cos \phi \sin \theta \sin \psi) \end{cases} \Leftarrow \vec{\omega}^r = \vec{\omega} - R_{BL} \begin{bmatrix} 0 \\ \omega_0 \\ 0 \end{bmatrix}$$

© ~~On-orbit Relative Equilibrium & Linearization~~

$$\begin{cases} M_x^G = -3\omega_0^2 (I_y - I_z) \sin \phi \cos \phi \cos^2 \theta \\ M_y^G = 3\omega_0^2 (I_z - I_x) \cos \phi \sin \theta \cos \theta \\ M_z^G = 3\omega_0^2 (I_x - I_y) \sin \phi \sin \theta \cos \theta \end{cases}$$

© On-orbit Relative Equilibrium ~~& Linearization~~

(Torque-free Motion $\Leftrightarrow M_x = M_y = M_z = 0$)

~~1) Relative Equilibrium~~

$$\begin{cases} \omega_x = 0, \quad \omega_y = \omega_0, \quad \omega_z = 0 \\ \psi = 0, \quad \theta = 0, \quad \phi = 0 \end{cases}$$

$\Rightarrow$ In a relative equilibrium, S/C body axes are aligned with LVLH axes ($\vec{i}_b = \vec{i}_L, \vec{j}_b = \vec{j}_L, \vec{k}_b = \vec{k}_L$)

© Linearization

$$\begin{cases} \omega_x = 0 + \Delta \omega_x \\ \omega_y = \omega_0 + \Delta \omega_y \\ \omega_z = 0 + \Delta \omega_z \end{cases}$$

$$\begin{cases} \psi = 0 + \Delta \psi \\ \theta = 0 + \Delta \theta \\ \phi = 0 + \Delta \phi \end{cases}$$

1) Linearized Pitch Dynamics

$$\begin{cases} \Delta \dot{\theta} = \Delta \omega_y \\ \Delta \dot{\omega}_y = \frac{3\omega_0^2(-I_x + I_z)}{I_y} \Delta \theta + \frac{I_x}{I_y} M_y \end{cases}$$

$$\Leftrightarrow I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + I_x M_y$$

Stable if $I_x > I_z$

II) Linearized Roll & Yaw Dynamics

$$\begin{cases} \Delta \dot{\psi} = \Delta \omega_z + \omega_0 \Delta \phi \\ \Delta \dot{\phi} = \Delta \omega_x - \omega_0 \Delta \psi \\ I_x \Delta \dot{\omega}_x = \omega_0 (I_y - I_z) \Delta \omega_z + 3\omega_0^2 (I_z - I_y) \Delta \phi + I_x M_x \\ I_z \Delta \dot{\omega}_z = \omega_0 (I_x - I_y) \Delta \omega_x + I_z M_z \end{cases}$$

$$\Leftrightarrow \begin{bmatrix} I_x & 0 \\ 0 & I_z \end{bmatrix} \begin{bmatrix} \Delta \ddot{\phi} \\ \Delta \ddot{\psi} \end{bmatrix} + \begin{bmatrix} -(I_x - I_y + I_z) & 0 \\ 0 & (I_z - I_x - I_y) \end{bmatrix} \omega_0 \begin{bmatrix} \Delta \phi \\ \Delta \psi \end{bmatrix} + \omega_0^2 \begin{bmatrix} 4(I_y - I_z) & 0 \\ 0 & I_y - I_x \end{bmatrix} \begin{bmatrix} \Delta \phi \\ \Delta \psi \end{bmatrix} = 0$$

Char. Eqn.

$$I_x I_z s^4 + I_x I_z (1 + 3k_1 + k_1 k_3) \omega_0^2 s^2 + I_x I_z (4k_1 k_3) \omega_0^4 = 0$$

$$k_1 = (I_y - I_x) / I_x$$

$$k_3 = (I_y - I_z) / I_z$$

Roll / Yaw Axis Stability Criteria: 4 poles on the imaginary axis.

$$\textcircled{1} \begin{cases} 1 + 3k_1 + k_1 k_3 > 0 \\ k_1 k_3 > 0 \\ (1 + 3k_1 + k_1 k_3)^2 - 16k_1 k_3 > 0 \end{cases}$$

Nec. & Sufficient conditions
(from Routh-~~Hurwitz~~ Hurwitz Criterion)

$$\textcircled{2} I_y > I_x > I_z$$

(Sufficient conditions
NOT Necessary conditions.)

$$\begin{aligned} k_3 &= (I_y - I_z) / I_z \\ k_1 &= (I_y - I_x) / I_x \\ b_2 &= I_x I_z (4k_1 k_3) \omega_0^4 \\ b_1 &= I_x I_z (1 + 3k_1 + k_1 k_3) \omega_0^2 \\ b_0 &= I_x I_z \end{aligned}$$

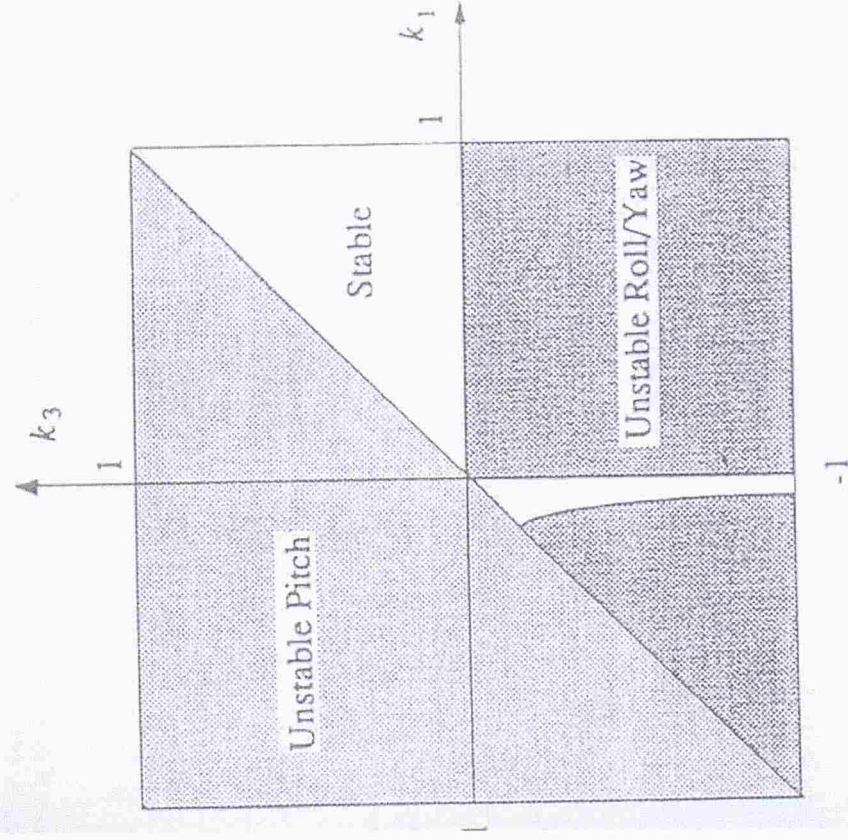
$$b_0 s^4 + b_1 s^2 + b_2 = 0$$

Char. Eqn.

Rigid S/C in a Circular Orbit: Stability

SAT 2011

- Gravity-gradient Stability Plot



<S/C ATTITUDE CONTROL - Open Loop>

⑥ Single Axis Maneuver

i) Assumptions

- a) rotational dynamics about a single principal axis
- b) ignore all gravity gradient and on-orbit effects

$$\Rightarrow I \ddot{\theta} = M$$

ii) Rest-to-Rest slewing maneuver

$$\begin{cases} \theta(0) = \theta_0, & \dot{\theta}(0) = 0 \\ \theta(t_f) = 0, & \dot{\theta}(t_f) = 0 \end{cases}$$

iii) Impulsive Control (need 2 impulses)

$$\ddot{\theta}(t) \neq 0$$

$$t = 0 \quad \theta(0) = \theta_0, \quad \dot{\theta}(0) = 0$$

$$t = 0^+ \quad \theta(0^+) = \theta_0, \quad \dot{\theta}(0^+) = 0 + \underbrace{\Delta \dot{\theta}(0)}_{\text{TBD}}$$

$$t \in [0^+, t_f^-]$$

$$\ddot{\theta}(t) = \Delta \dot{\theta}(0)$$

$$\dot{\theta}(t) = \Delta \dot{\theta}(0)t + \dot{\theta}(0)$$

$$t = t_f$$

$$\theta(t_f) = 0 = \Delta \dot{\theta}(0)t_f + \theta_0 \Rightarrow \Delta \dot{\theta}(0) = -\theta_0/t_f$$

$$\dot{\theta}(t_f) = 0 = \dot{\theta}(t_f^-) + \Delta \dot{\theta}(t_f^-)$$

$$= \Delta \dot{\theta}(0) + \Delta \dot{\theta}(t_f^-) \neq \Delta \dot{\theta}(0)$$

$$\Rightarrow \Delta \dot{\theta}(t_f^-) = -\Delta \dot{\theta}(0) = \theta_0/t_f$$

$$t = t_f$$

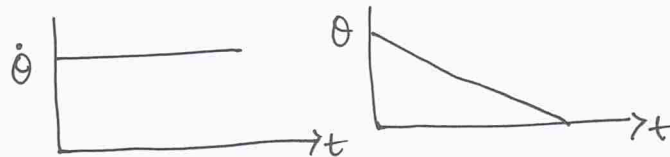
$$\theta(t_f) = 0 = \Delta \dot{\theta}(0) t_f + \theta_0$$

$$\Rightarrow \underline{\Delta \dot{\theta}(0) = -\theta_0 / t_f}$$

$$\begin{aligned} \dot{\theta}(t_f) = 0 &= \dot{\theta}(t_f^-) + \Delta \dot{\theta}(t_f) \\ &= \Delta \dot{\theta}(0) + \Delta \dot{\theta}(t_f^-) \end{aligned}$$

$$\Rightarrow \underline{\Delta \dot{\theta}(t_f^-) = -\Delta \dot{\theta}(0) = \theta_0 / t_f}$$

$$\therefore \text{Total } |\Delta \dot{\theta}| = \frac{2|\theta_0|}{t_f}$$



$\theta = \dot{\theta} = 0$ is an equilibrium

IV) Piecewise Constant Maneuver

$$\text{Assume } M(t) = \begin{cases} \bar{M} & 0 \leq t \leq t_s \\ -\bar{M} & t_s \leq t \leq t_f \end{cases}$$

$$(a) \quad 0 \leq t \leq t_s \Rightarrow \ddot{\theta} = \bar{M}/I$$

$$\dot{\theta}(t) = \frac{\bar{M}}{I} t$$

$$\theta(t) = \frac{\bar{M}}{I} \frac{t^2}{2} + \theta_0$$

$$(b) \quad t = t_s$$

$$\dot{\theta}(t_s) = \frac{\bar{M}}{I} t_s$$

$$\theta(t_s) = \frac{\bar{M}}{I} \frac{t_s^2}{2} + \theta_0$$

$$(c) \quad t_s \leq t \leq t_f \Rightarrow \ddot{\theta} = -\bar{M}/I$$

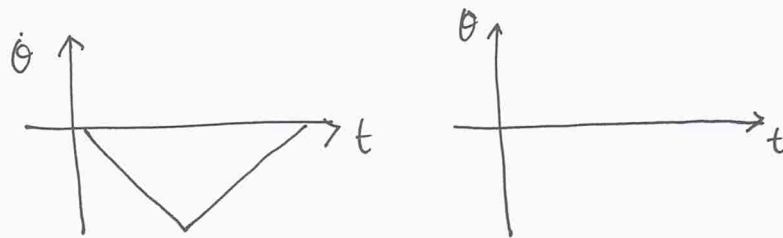
$$\dot{\theta}(t) = -\frac{\bar{M}}{I} (t - t_s) + \frac{\bar{M}}{I} t_s$$

$$\theta(t) = -\frac{\bar{M}}{I} \frac{(t - t_s)^2}{2} + \frac{\bar{M}}{I} t_s (t - t_s) + \frac{\bar{M}}{I} \frac{t_s^2}{2} + \theta_0$$

Introducing Boundary Conditions.

$$\begin{cases} \dot{\theta}(t_f) = 0 = -\frac{\bar{M}}{I}(t_f - t_s) + \frac{\bar{M}}{I}t_s \\ \theta(t_f) = 0 = -\frac{\bar{M}}{I}(t_f - t_s)^2 + \frac{\bar{M}}{I}t_s(t_f - t_s) + \frac{\bar{M}}{I}\frac{t_s^2}{2} + \theta_0 \end{cases}$$

$$\Rightarrow \begin{cases} t_s = \frac{t_f}{2} \\ \bar{M} = -4I\theta_0/t_f^2 \end{cases}$$



V) Time - Optimal Control

$$J = \int_0^{t_f} M_m dt = M_m t_f$$

$$\text{s.t. } I\ddot{\theta} = M, \quad |M| \leq M_{\max}$$

$$\theta(0) = \theta_0, \quad \dot{\theta}(0) = 0$$

$$\theta(t_f) = 0, \quad \dot{\theta}(t_f) = 0$$

It turns out that

$$t_f^* = \sqrt{\frac{4I\theta_0}{M_{\max}}}$$

(Please derive by yourself.)

12/1 ① 3D Rest-to-Rest Slewing Maneuver (Stationary Case) (3/)

$$t = 0 \rightarrow t = t_f$$

$$R(0) = R_0$$

$$R(t_f) = I$$

$$\omega(0) = 0$$

$$\omega(t_f) = 0$$

Assume that $R_0 = e^{S(\vec{a})\theta_0}$, $|\vec{a}|=1$ expressed in exponential coordinates

Consider a maneuver with

$$R(t) = e^{S(\vec{a})\theta(t)}, \quad \theta(0) = \theta_0, \quad \theta(t_f) = 0 \quad (1)$$

$$\begin{aligned} \dot{R}(t) &= S(\vec{a})\dot{\theta}(t) e^{S(\vec{a})\theta(t)} \leftarrow \text{direct differentiation} \\ \dot{R}(t) &= S(\vec{\omega})R(t) = S(\vec{\omega})e^{S(\vec{a})\theta(t)} \leftarrow \text{attitude kinematic eqn.} \end{aligned}$$

$$\Rightarrow S(\vec{a})\dot{\theta}(t) = S(\vec{\omega})$$

$$S(\vec{a})\dot{\theta}(t) = S(\vec{\omega})$$

$$\Rightarrow \vec{\omega} = \vec{a}\dot{\theta}, \quad \dot{\theta}(0) = 0, \quad \dot{\theta}(t_f) = 0 \quad (2)$$

Choose $\theta(t)$, $0 \leq t \leq t_f$

$$\text{to satisfy } \begin{pmatrix} \theta(0) = \theta_0 & \dot{\theta}(0) = 0 \\ \theta(t_f) = 0 & \dot{\theta}(t_f) = 0 \end{pmatrix} \quad (3)$$

Assume

$$\theta(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3 \quad (4)$$

Introduce (3) $\rightarrow$ (4),

$$b_0 = \theta_0, \quad b_1 = 0, \quad b_2 = -\frac{3\theta_0}{t_f^2}, \quad b_3 = \frac{2\theta_0}{t_f^3}$$

$$\Rightarrow \theta(t) = \theta_0 - 3\theta_0\left(\frac{t}{t_f}\right)^2 + 2\theta_0\left(\frac{t}{t_f}\right)^3$$

Now, from Euler's EOM.

$$I\dot{\omega} + \omega \times I\omega = M$$

$$\Rightarrow I\ddot{\theta} + \dot{\theta} \times I\dot{\theta} = M$$

$$\Rightarrow M = T\ddot{\theta} + \dot{\theta}^2(\vec{a} \times I\vec{a})$$

3-D Rest-to-Rest Slewing Maneuver (On-Orbit Case)

1) Pitch Dynamics

$$\begin{cases} \Delta \dot{\theta} = \Delta \omega_y \\ \Delta \dot{\omega}_y = \frac{3\omega_0^2(-I_x + I_z)}{I_y} \Delta \theta + \frac{M_y}{I_y} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \Delta \dot{\theta} \\ \Delta \dot{\omega}_y \end{bmatrix} = A \begin{bmatrix} \Delta \theta \\ \Delta \omega_y \end{bmatrix} + B M_y, \quad \begin{bmatrix} \Delta \theta \\ \Delta \omega_y \end{bmatrix}_{t=0} = \begin{bmatrix} \Delta \theta_0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \Delta \theta \\ \Delta \omega_y \end{bmatrix}_{t=t_f} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Two-Impulse Solution

$$\begin{bmatrix} \Delta \theta(t_f^-) \\ \Delta \omega_y(t_f^-) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \theta_0 \\ \Delta \omega_y(0^+) \end{bmatrix} = \begin{bmatrix} 0 \\ \Delta \omega_y(t_f^-) \end{bmatrix}$$

$$\begin{cases} \Delta \omega_y(0^+) = -\frac{e_1^T e^{At_f} e_1}{e_1^T e^{At_f} e_2} \Delta \theta_0, \\ \Delta \omega_y(t_f^-) = e_2^T e^{At_f} e_1 \Delta \theta_0 - e_2^T e^{At_f} e_2 \left(\frac{e_1^T e^{At_f} e_1}{e_1^T e^{At_f} e_2} \right) \Delta \theta_0 \\ e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{cases}$$

Piecewise Constant Maneuver

$$M_y = \begin{cases} \bar{M}_y & 0 \leq t \leq t_s \\ -\bar{M}_y & t_s \leq t \leq t_f \end{cases}$$

$$\begin{aligned} \begin{bmatrix} \Delta \theta(t_f) \\ \Delta \omega_y(t_f) \end{bmatrix} &= e^{At_f} \begin{bmatrix} \Delta \theta_0 \\ 0 \end{bmatrix} + \int_0^{t_s} e^{A(t_f-s)} B(\bar{M}_y) ds + \int_{t_s}^{t_f} e^{A(t_f-s)} B(-\bar{M}_y) ds \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow 2 \text{ eqns for } (t_s, \bar{M}) \end{aligned}$$

11) Roll / Yaw Dynamics

$$\begin{bmatrix} \Delta \dot{\psi} \\ \Delta \dot{\phi} \\ \Delta \dot{w}_x \\ \Delta \dot{w}_z \end{bmatrix} = A \begin{bmatrix} \Delta \psi \\ \Delta \phi \\ \Delta w_x \\ \Delta w_z \end{bmatrix} + B \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$

$$\begin{bmatrix} \Delta \psi(0) \\ \Delta \phi(0) \\ \Delta w_x(0) \\ \Delta w_z(0) \end{bmatrix} = \begin{bmatrix} \Delta \psi_0 \\ \Delta \phi_0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \Delta \psi(t_f) \\ \Delta \phi(t_f) \\ \Delta w_x(t_f) \\ \Delta w_z(t_f) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Two-Impulse Solution

$$\begin{bmatrix} \Delta \psi(t_f) \\ \Delta \phi(t_f) \\ \Delta w_x(t_f) \\ \Delta w_z(t_f) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \psi_0 \\ \Delta \phi_0 \\ \Delta w_x(0^+) \\ \Delta w_z(0^+) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \Delta w_x(t_f^-) \\ \Delta w_z(t_f^-) \end{bmatrix}$$

Piecewise Constant Solution

$$M_x = \begin{cases} \bar{M}_x, & 0 \leq t \leq t_s^x \\ -\bar{M}_x, & t_s^x \leq t \leq t_f \end{cases}$$

$$M_z = \begin{cases} \bar{M}_z, & 0 \leq t \leq t_s^z \\ -\bar{M}_z, & t_s^z \leq t \leq t_f \end{cases}$$

$$\begin{bmatrix} \Delta \psi(t_f) \\ \Delta \phi(t_f) \\ \Delta w_x(t_f) \\ \Delta w_z(t_f) \end{bmatrix} = e^{At_f} \begin{bmatrix} \Delta \psi_0 \\ \Delta \phi_0 \\ 0 \\ 0 \end{bmatrix} + \int_0^{t_s^x} e^{A(t_f-s)} B_1(\bar{M}_x) ds + \int_{t_s^x}^{t_f} e^{A(t_f-s)} B_1(-\bar{M}_x) ds \\ + \int_0^{t_s^z} e^{A(t_f-s)} B_2(\bar{M}_z) ds + \int_{t_s^z}^{t_f} e^{A(t_f-s)} B_2(-\bar{M}_z) ds$$

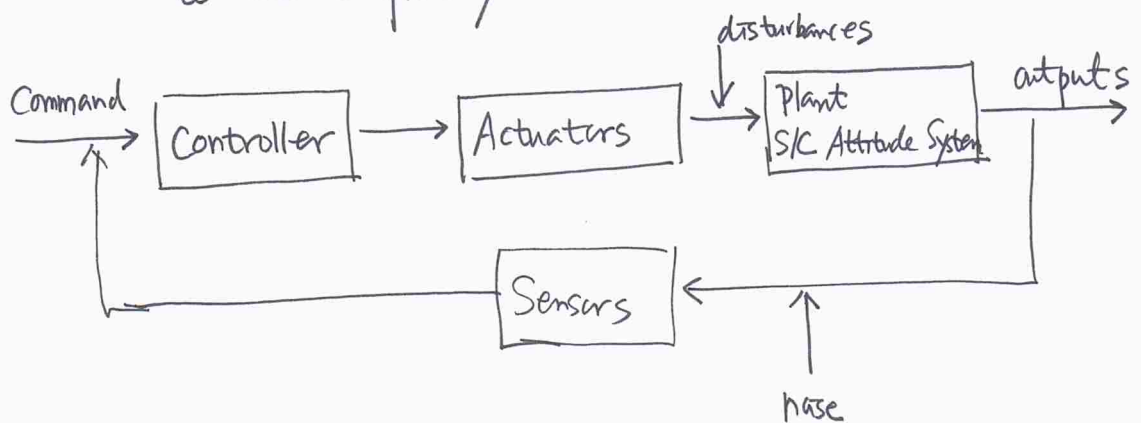
$$= 0 \Rightarrow 4 \text{ eqn for } 4 \text{ unknowns } (\bar{M}_x, \bar{M}_z, t_s^x, t_s^z)$$

Lemma) If A is invertible, then
 $\int_0^t e^{A(t-s)} B ds = A^{-1} [e^{At} - e^{A \cdot 0}] B$

< SIC ATTITUDE CONTROL & STABILIZATION - CLOSED LOOP >

① Attitude Control System

Composed of Plant, Sensors, and Actuators which are assumed to be completely known



② Control of Angular Velocity Vector (only!)

$$I\dot{\vec{\omega}} + \vec{\omega} \times I\vec{\omega} = \vec{M}$$

Control objective : $\vec{\omega}_0 \rightarrow \vec{\omega} = 0$

†) Linear Control

$$I\dot{\vec{\omega}} = \vec{M} \Rightarrow \begin{cases} I_x \dot{\omega}_x = M_x \\ I_y \dot{\omega}_y = M_y \\ I_z \dot{\omega}_z = M_z \end{cases} \quad M_x = -K(\omega_x - \omega_x^c)$$

$$\Rightarrow I_x \dot{\omega}_x + K(\omega_x - \omega_x^c) = 0$$

$$\Rightarrow \left(\text{If } \omega_x^c = 0 \text{ \& } K > 0, \right. \\ \left. \text{then } \omega_x \rightarrow 0 \text{ for } \forall \omega_{x0} \neq 0 \right)$$

ii) Nonlinear Control

$$\begin{pmatrix} I\dot{\vec{\omega}} + \vec{\omega} \times I\vec{\omega} = \vec{M} \end{pmatrix} \quad (1)$$

$$\begin{pmatrix} \vec{M} = \vec{\omega} \times I\vec{\omega} - k(\vec{\omega} - \vec{\omega}^c) \end{pmatrix} \quad (2)$$

$$\Rightarrow I\dot{\vec{\omega}} + k(\vec{\omega} - \vec{\omega}^c) = 0 \Leftarrow (2) \rightarrow (1)$$

$\Rightarrow$ Choose $k > 0$ s.t. the closed-loop system is stable

① S/C Attitude Stabilization (Dynamics + Kinematics)

I) Linear On-orbit Control

Pitch Control Problem

$$\begin{bmatrix} \Delta\dot{\theta} \\ \Delta\dot{\omega}_y \end{bmatrix} = \begin{bmatrix} \Delta\omega_y \\ \frac{3\omega_0^2(-I_x + I_z)}{I_y} \Delta\theta \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{M_y}{I_y} \end{bmatrix}$$

$$M_y = -k_p(\Delta\theta - \Delta\theta^c) - k_d\Delta\omega_y$$

May use LQR to choose a pair of (k_p, k_d) s.t.
the closed-loop is stable

~~i) Nonlinear Control I: Feedback Linearization~~
~~= Exact Linearization~~

ii) Nonlinear Control by Feedback Linearization
= Exact Linearization

Nonlinear System $\dot{x} = f(x, u)$ goes through

1) variable change

2) feedback transformation $\dot{z} = Az + Bv$

Case I: Nonlinear Attitude Control ✓ with Euler Angles

$$\left\{ \begin{aligned} \dot{w}_x &= \frac{I_y - I_z}{I_x} w_y w_z + \frac{u_x}{I_x} \\ \dot{w}_y &= \frac{I_z - I_x}{I_y} w_z w_x + \frac{u_y}{I_y} \\ \dot{w}_z &= \frac{I_x - I_y}{I_z} w_x w_y + \frac{u_z}{I_z} \\ \dot{\psi} &= (w_y \sin \phi + w_z \cos \phi) \sec \theta \\ \dot{\theta} &= w_y \cos \phi - w_z \sin \phi \\ \dot{\phi} &= w_x + w_y \sin \phi \tan \theta + w_z \cos \phi \tan \theta \end{aligned} \right.$$

Control Objective

- ① Asymptotically Stable Closed Loop
- ② Closed-loop equilibrium at $\psi = \psi_d$, $\theta = \theta_d$, $\phi = \phi_d$
- ③ Other benefits of feedback

$$\left\{ \begin{aligned} \ddot{\psi} &= (\dot{w}_y \sin \phi + w_z \cos \phi) \sec \theta + (w_y \cos \phi - w_z \sin \phi) \dot{\phi} \sec \theta \\ &\quad - \cancel{(w_y \sin \phi)} + (w_y \sin \phi + w_z \cos \phi) \frac{\sin \theta}{\cos^2 \theta} \dot{\theta} \\ &\quad \vdots \end{aligned} \right.$$

$$\Rightarrow \left\{ \begin{aligned} \ddot{\psi} &= \left(\frac{u_y}{I_y} \sin \phi + \frac{u_z}{I_z} \cos \phi \right) \sec \theta + f_\psi \\ \ddot{\theta} &= \frac{u_y}{I_y} \cos \phi - \frac{u_z}{I_z} \sin \phi + f_\theta \\ \ddot{\phi} &= \frac{u_x}{I_x} + \frac{u_y}{I_y} \sin \phi \tan \theta + \frac{u_z}{I_z} \cos \phi \tan \theta + f_\phi \end{aligned} \right.$$

Define

$$\begin{cases} V_\psi = \ddot{\psi} & \text{yaw channel} \\ V_\theta = \ddot{\theta} & \text{pitch channel} \\ V_\phi = \ddot{\phi} & \text{roll channel} \end{cases}$$

Let

$$\begin{cases} V_\psi = -C_\psi \dot{\psi} - k_\psi (\psi - \psi_d) \\ V_\theta = -C_\theta \dot{\theta} - k_\theta (\theta - \theta_d) \\ V_\phi = -C_\phi \dot{\phi} - k_\phi (\phi - \phi_d) \end{cases}$$

If $C > 0$ and $C^2 - 4k < 0$

then the closed-loop is 'Globally' Asymptotically Stable (GAS)

and as $t \rightarrow \infty$, $(\psi \rightarrow \psi_d, \theta \rightarrow \theta_d, \phi \rightarrow \phi_d)$

$$\begin{bmatrix} V_\phi \\ V_\theta \\ V_\psi \end{bmatrix} = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \begin{bmatrix} \frac{u_x}{I_x} \\ \frac{u_y}{I_y} \\ \frac{u_z}{I_z} \end{bmatrix} + \begin{bmatrix} f_\phi \\ f_\theta \\ f_\psi \end{bmatrix} = \begin{bmatrix} -C_\phi \dot{\phi} - k_\phi (\phi - \phi_d) - f_\phi \\ -C_\theta \dot{\theta} - k_\theta (\theta - \theta_d) - f_\theta \\ -C_\psi \dot{\psi} - k_\psi (\psi - \psi_d) - f_\psi \end{bmatrix}$$

$$\begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} I_x & 0 \\ 0 & I_y \\ 0 & I_z \end{bmatrix} \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix}^{-1} \begin{bmatrix} -C_\phi \dot{\phi} - k_\phi (\phi - \phi_d) - f_\phi \\ -C_\theta \dot{\theta} - k_\theta (\theta - \theta_d) - f_\theta \\ -C_\psi \dot{\psi} - k_\psi (\psi - \psi_d) - f_\psi \end{bmatrix}$$

~~exists if $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$~~

exists if $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

Case II: Nonlinear Attitude Control & Stabilization with Quaternion

$$\begin{cases} I \dot{w} = -S(w) I w + M \\ \dot{q} = \frac{1}{2} S(w) q + \frac{1}{2} q_4 w \\ \dot{q}_4 = -\frac{1}{2} w^T q \end{cases}$$

$$\begin{aligned} \Rightarrow \ddot{q} &= \frac{1}{2} S(\dot{w}) q + \frac{1}{2} S(w) \dot{q} + \frac{1}{2} \dot{q}_4 w + \frac{1}{2} q_4 \dot{w} \\ &= -\frac{1}{2} S(q) \dot{w} + \frac{1}{2} S(w) \left[\frac{1}{2} S(w) q + \frac{1}{2} q_4 w \right] - \frac{1}{4} (w^T q) w + \frac{1}{2} q_4 \dot{w} \\ &= \frac{1}{2} [q_4 I_{3 \times 3} - S(q)] \dot{w} + \frac{1}{4} S^2(w) q + \frac{1}{4} S(w) q_4 w - \frac{1}{4} (w^T q) w \end{aligned}$$

Using $S^2(w) q = (w w^T) q$, $S(w) q_4 w = -(w^T w) q$, $(w^T q) w = (w w^T) q$

$$\begin{aligned} \Rightarrow \ddot{q} &= \frac{1}{2} [q_4 I_{3 \times 3} - S(q)] \dot{w} + \frac{1}{4} [\cancel{w w^T} - w^T w I_{3 \times 3}] q - \frac{1}{4} \cancel{(w w^T) q} \\ &= \frac{1}{2} [q_4 I_{3 \times 3} - S(q)] \dot{w} - \frac{1}{4} (w^T w) q \end{aligned}$$

Introducing $\dot{w} = I^{-1} [-S(w) I w + M]$

$$\begin{aligned} \Rightarrow \ddot{q} &= -\frac{1}{2} [q_4 I_{3 \times 3} - S(q)] I^{-1} S(w) I w - \frac{1}{4} (w^T w) q \\ &\quad + \frac{1}{2} [q_4 I_{3 \times 3} - S(q)] I^{-1} M \end{aligned}$$

Let ~~$\ddot{q}$~~

$$\ddot{q} = v = -K_1 (q - q_c) - K_2 \dot{q}$$

$$\begin{aligned} M &= S(w) I w + I [q_4 I_{3 \times 3} - S(q)]^T [-2k_1 (q - q_c) \\ &\quad - 2k_2 \dot{q} + \frac{1}{2} (w^T w) q] \end{aligned}$$

Lemma) $[q_4 I_{3 \times 3} - S(q)]$ is invertible iff $q_4 \neq 0$, i.e., $q^T q < 1$
and

$$[q_4 I_{3 \times 3} - S(q)]^{-1} = \frac{1}{q_4} [q_4^2 I_{3 \times 3} + q q^T + q_4 S(q)]$$

Pf) $\frac{1}{q_4} [q_4 I_{3 \times 3} + q q^T + q_4 S(q)] [q_4 I_{3 \times 3} - S(q)] = \dots = I_{3 \times 3}$

(ii) Nonlinear Stabilization by Lyapunov Approach.

$$I \dot{w} = -S(w)Iw + M$$

$$\dot{q} = \frac{1}{2} S(w)q + \frac{1}{2} q_4 w$$

$$\dot{q}_4 = -\frac{1}{2} w^T q$$

Thm) Consider

$$M = S(\bar{w})Iw + I(-k(q - q_c) - Dw)$$

where $k > 0$ is a scalar gain

and $D > 0$ is a positive definite matrix.

Then the closed-loop is asymptotically stable

and as $t \rightarrow \infty$, $q \rightarrow q_c$ and $\bar{w} \rightarrow 0$

Pf) Without loss of generality, let $q_c = 0$

The closed-loop equations are

$$\begin{cases} \dot{w} = -kq - Dw \\ \dot{q} = \frac{1}{2} S(w)q + \frac{1}{2} q_4 w \\ \dot{q}_4 = -\frac{1}{2} w^T q \end{cases}$$

Consider a candidate Lyapunov function

$$V = 2k [1 - \sqrt{1 - \mathbf{q}^T \mathbf{q}}] + \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

$$= 2k (1 - q_4) + \frac{1}{2} \mathbf{w}^T \mathbf{w}$$

i) $V = 0$ at equilibrium and $V > 0$ otherwise

$$\begin{aligned} \text{ii) } \dot{V} &= -2k \dot{q}_4 + \mathbf{w}^T \dot{\mathbf{w}} \\ &= -2k \left(-\frac{1}{2} \mathbf{w}^T \mathbf{q} \right) + \mathbf{w}^T (-k \mathbf{q} - D \mathbf{w}) \\ &= -\mathbf{w}^T D \mathbf{w} \leq 0 \end{aligned}$$

ii) Use La Salle's theorem to complete the proof of AS.

(Q.E.D.)

$$\text{Let } \mathbf{M} = S(\mathbf{w}) \mathbf{I} \mathbf{w} - k \mathbf{I} \mathbf{q} - d \mathbf{I} \dot{\mathbf{w}}$$

Then the closed-loop is

$$\begin{cases} \dot{\mathbf{w}} = -k \mathbf{q} - d \mathbf{w} \\ \dot{\mathbf{q}} = \frac{1}{2} S(\mathbf{w}) \mathbf{q} + \frac{1}{2} \mathbf{q}_4 \mathbf{w} \\ \dot{q}_4 = -\frac{1}{2} \mathbf{w}^T \mathbf{q} \end{cases}$$

We want the closed-loop responses to be rotations about a fixed axis with $\mathbf{q}(0) = \mathbf{q}_0$, $\mathbf{w}(0) = 0$.

$$\text{Let } \begin{pmatrix} \mathbf{q}(t) \\ \mathbf{w}(t) \end{pmatrix} = \begin{pmatrix} \sin(\frac{\theta}{2}) \mathbf{a} \\ \dot{\theta} \mathbf{a} \end{pmatrix} \Leftrightarrow (\text{Quaternion} \leftrightarrow \text{Axis-Angle Variable})$$

Then $\ddot{\theta} a = -k \sin\left(\frac{\theta}{2}\right) a - d \dot{\theta} a$

$$\Rightarrow \ddot{\theta} + d \dot{\theta} + k \sin\left(\frac{\theta}{2}\right) = 0$$

For small angles,

$$\ddot{\theta} + d \dot{\theta} + \frac{k}{2} \theta = 0$$

Choose $d > 0$ and $d^2 - 2k < 0$

s.t. the closed-loop system is AS (for small angle maneuvers)